Van Phan

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Education

Georgia State University — *Atlanta, GA*

B.S. Mathematics, GPA: 4.12/4.30 — Expected Graduation: May. 2027

- *Relevant Coursework*: Linear Algebra, Differential Equations, Mathematical Biology, Biostatistics, Optimization, Numerical Analysis, Analysis, Modern Algebra, Probability and Statistics, Applied Combinatorics, Calculus, Computer Science

Research Projects

Density and BI-RADS - Aware Triage and Automatic Report Generation for Mammography

- Developed the **Density and BI-RADS-Aware Triage and Report Generation (DB-ATRG)** framework to automate diagnostic text generation and enable risk-based case prioritization, flagging scans with **extremely dense breasts (ACR Category D)** for supplemental screening.
- Fine-tuned the **4-billion parameter MedGemma 1.5** vision-language model using **QLoRA** on the **DMID and VinDr-Mammo dataset**, achieving an **ACR Density Accuracy of 0.7039**, a **ROUGE-L score of 0.8650**, and a **METEOR score of 0.9001** for clinical report generation.
- Accelerated clinical review in simulations on a 100-case cohort by applying a **Cumulative Urgency Score**, capturing **100% of high-risk malignancies** (BI-RADS 4 and 5) within the first 20% of the reading workload (compared to 40% in FIFO) and shifting the **mean rank of severe cases from 42.8 down to 3**.

Modeling Wound Healing Without Migration and Proliferation

- Established a mathematical framework for Age-related Macular Degeneration (AMD) therapeutics, addressing a leading cause of vision loss by modeling Retinal Pigment Epithelium (RPE) wound healing.
- Simulated purse-string contraction and cell fusion repair mechanisms in Python-based CompuCell3D, by utilizing a **Q-Potts cellular model** and **Monte Carlo evaluation** of the **Hamiltonian free energy states**.
- Proved that **higher cellular volume constraints** yield **statistically faster wound closure**, quantifying intervention viability by analyzing texture tensors and spatial points across different algorithmic iterations.

Enhanced Physics-Informed Neural Networks for Collective Cancer Invasion

- Engineered a **Physics-Informed Neural Networks (PINNs)** model to simulate collective cancer invasion, resulting in a data-efficient, mesh-free solution for complex **Partial Differential Equations (PDEs)**.
- Developed a framework utilizing **Tensorized Fourier Neural Operators (TFNO)** and **PyTorch's autograd engine** to encode physical laws (PDEs, boundary, and initial conditions) directly into the model's loss function, reducing reliance on labeled data.
- Explored advanced techniques, including a **sequence-to-sequence (Seq2seq) PINO** approach and **Augmented Lagrangian methods (ALM)**, to overcome model failure modes like vanishing gradients and improve solution accuracy and stability.

Research Paper

- **[PrePrint] DB-ATRG: The Density and BI-RADS-Aware Triage and Automatic Report Generation System for Mammography**, doi: 10.64898/2026.07.22.26358655.

Posters & Presentations

Accelerating Breast Cancer Diagnosis: AI-Driven Triage and Prioritization in Mammography,
- Poster presented at 2026 SIAM Conference on Mathematics of Data Science.

Density and BI-RADS - Aware Triage and Automatic Report Generation System for Mammography,
- Poster presented at Georgia State Undergraduate Research Conference, 2026.

Enhanced Physics-Informed Neural Networks for Collective Cancer Invasion,
- **Best Project Award** at Summer Undergraduate Research Symposium, 2025.

Morphological Feature Analysis of Retinal Pigment Epithelial Cell from Mice during Aging,
- Poster presented at Georgia State Undergraduate Research Conference, 2025.

Modeling Wound Healing Without Migration and Proliferation,
- Poster presented at Summer Undergraduate Research Symposium, 2024.

Research and Work Experience

RIMMES Program — Sep. 2025 – Present
Undergraduate Researcher (Advisor: Dr. Russell Jeter) — *Atlanta, GA*

- Led a 3-member student team in DB-ATRG project, developing a system to automatically report clinical report and prioritize high-risk cases in mammography.

Georgia State University — Aug. 2025 – Present
Chemistry Laboratory Teaching Assistant — *Atlanta, GA*

- Mentored 50+ undergraduate students in Survey of Chemistry I and II, improving understanding of fundamental principles and reducing procedural errors through one-on-one guidance and demonstration of proper laboratory techniques.

Center for the Advancement of Students and Alumni — Jun. 2024 – August. 2025
Undergraduate Researcher (Advisor: Dr. Yi Jiang) — *Atlanta, GA*

- Led a 3-member student team in two concurrent computational biology projects, developing wound healing simulations and morphological analysis of Retinal Pigment Epithelial (RPE) cells.

Near-peer Mentor

- Provided near-peer mentorship and research support to students in the Math Path Program, guiding research technique implementation and project development for the Enhanced Physics-Informed Neural Networks for Collective Cancer Invasion project.

Selected Projects

Morphological Feature Analysis of Retinal Pigment Epithelial Cell from Mice during Aging

- Developed an automated end-to-end machine learning pipeline for classifying Retinal Pigment Epithelial (RPE) cell during aging, generating insights for potential therapeutic strategies in Age-related Macular Degeneration.
- Achieved highly accurate classification as measured by **90%+ cross-validation F1-score**, by extracting **133 morphological and texture-based features** from 326 RPE cell images and implementing a **stacking ensemble** of **XGBoost, LightGBM, and CatBoost** with a Logistic Regression meta-learner.
- Identified critical RPE aging biomarkers, as demonstrated by top feature importance rankings, suggesting **Local Binary Patterns (LBP)** and **geometric shape metrics** (circularity, aspect ratio) as the most influential classification variables.

Honors & Awards

- **Best Project Award** at Summer Undergraduate Research Symposium (2025)
- **President's List** at Georgia State University (2023, 2024, 2025, 2026)
- **Prospect Award** at Vietnam Science and Engineering Fair (ViSEF) (2022), National Level
- **First Place** at Vietnam Science and Engineering Fair (ViSEF) (2022), City Level
- **Consolation Prize** at Engineering and Science Fair of Ho Chi Minh City University of Technology (2022)
- **Bronze Prize** in Physics at the Ho Chi Minh City Excellent Student Contest (2019)

Technical Skills

- **Programming & Tools**: Python, Julia, R, MATLAB, SQL, Git/GitHub, Linux/CLI, LaTeX, CompuCell3D, Microsoft Office Suite
- **Scientific & Math Modeling**: Physics-Informed Neural Networks, PDE Modeling, Tensorized Fourier Neural Operators, Augmented Lagrangian Method, Q-Potts Cellular Modeling, Monte Carlo Evaluation
- **ML, LLM & AI**: LLMs, Fine-Tuning, Quantization, Transformers, TRL, Hugging Face, Ensemble Methods
- **Libraries**: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, SciPy, Scikit-image, Pillow, Mahotas, Matplotlib, Seaborn